

Spivak Chapter 10 Problems

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1 Problem 1

Find $f'(x)$ for each of the following f .

1. $f(x) = \sin(x + x^2).$
 $(1 + 2x) \cos(x + x^2).$

2. $f(x) = \sin x + \sin x^2.$
 $\cos x + 2x \cos x^2.$

3. $f(x) = \sin(\cos x).$
 $-\sin x \cdot \cos(\cos x).$

4. $f(x) = \sin(\sin x).$
 $\cos x \cdot \cos(\sin x).$

5.

$$f(x) = \sin\left(\frac{\cos x}{x}\right).$$

$$-\frac{x \sin x + \cos x}{x^2} \cdot \cos\left(\frac{\cos x}{x}\right).$$

6.

$$f(x) = \frac{\sin(\cos x)}{x}.$$

$$\frac{-x(\sin x \cos(\cos x)) - \sin(\cos x)}{x^2}.$$

7. $f(x) = \sin(x + \sin x).$
 $(1 + \cos x)(\cos(x + \sin x)).$

2 Problem 7

1. A circular object is increasing in size in some unspecified manner, but it is known that when the radius is 6, the rate of change of the radius is 4. Find the rate of change of the area when the radius is 6.

$$A = \pi r^2,$$

$$r' \cdot A'(r),$$

$$4 \cdot 2\pi r,$$

$$8\pi r = 48\pi.$$

2. Suppose we are now informed that the circular object we have been watching is really the cross section of a spherical object. Find the rate of change of the volume when the radius is 6.

$$V(r) = \frac{4}{3}\pi r^3,$$

$$r' \cdot V'(r),$$

$$4 \cdot 4\pi r^2,$$

$$576\pi.$$

3 Problem 9

Particle A moves along the positive horizontal axis, and particle B along the graph of $f(x) = -\sqrt{3}x, x \leq 0$. At a certain time, A is at the point (5,0) and moving with speed 3 units/sec; and B is at a distance of 3 units from the origin and moving with speed 4 units/sec. At what rate is the distance between A and B changing?

$$a(t) = 5; a'(t) = 3,$$

$$9 = b(t)^2 + 3b(t)^2; 4 = b'(t)^2 + 3b'(t)^2,$$

$$b(t) = -\frac{3}{2}, b'(t) = -2,$$

$$d(t)^2 = (a(t) - b(t))^2 + (3b(t))^2,$$

$$2d'(t)d(t) = 2(a(t) - b(t))(a'(t) - b'(t)) + 6b'(t)b(t),$$

$$d(t) = \sqrt{(5 + \frac{3}{2})^2 + 3(\frac{3}{2})^2} = 7,$$

$$d'(t) = \frac{(a(t)-b(t))(a'(t)-b'(t))+3b'(t)b(t)}{d(t)},$$

$$d'(t) = \frac{(5+\frac{3}{2})(3+2)+(3 \cdot -2 \cdot -\frac{3}{2})}{7} = \frac{83}{14}.$$

4 Problem 11

Find $f'(0)$ if

$$f(x) = \begin{cases} g(x) \sin \frac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

and

$$g(0) = g'(0) = 0.$$

$$\lim_{x \rightarrow 0} \frac{f(x) - 0}{x} = \lim_{x \rightarrow 0} \frac{g(x) \sin(\frac{1}{x})}{x},$$

$$\lim_{x \rightarrow 0} \frac{g(x)}{x} = g'(x) = 0,$$

$$|\sin(\frac{1}{x})| \leq 1.$$

Thus, $f'(0) = 0$.

5 Problem 16

1. Prove that if f is differentiable at a , then $|f|$ is also differentiable at a , provided that $f(a) \neq 0$.

Because f is differentiable at a , it is continuous at a . In the case of $f(a) > 0$, $|f|'(a) = f'(a)$. If $f(a) < 0$, then $|f|'(a) = -f'(a)$.

2. Prove that if f and g are differentiable at a , then the functions $\max(f, g)$ and $\min(f, g)$ are differentiable at a , provided that $f(a) \neq g(a)$.

Because f, g are differentiable at a , they are continuous at a . Then $\max(f, g)$ is either equal to f' or g' . Similar for $\min(f, g)$.

6 Problem 20

1. If $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, find a function g such that $g' = f$. Find another.

$$g(x) = a_{n+1} x^{n+1} + a_n x^n + \dots + a_0 x.$$

$$g(x) = \frac{a_n x^{n+1}}{n+1} + \frac{a_{n-1} x^n}{n} + \dots + \frac{a_1 x^2}{2} + a_0 x + c.$$

2. If

$$f(x) = \frac{b_2}{x^2} + \frac{b_3}{x^3} + \dots + \frac{b_m}{x^m},$$

find a function with $g' = f$.

$$g(x) = -\frac{b_2}{x} - \frac{b_3}{2x^2} - \dots - \frac{b_m}{(m-1)x^{m-1}}.$$